



"The learning companion "

www.ktunotes.in

Module - III

Ordinary Differential Equations

Syllabus

Text II

2.1, 2.2, 2.5, 2.6, 2.7, 2.10, 3.1, 3.2, 3.3

Differential Equations

It is an equation which involves differentials or derivatives.

eg: $x dy + y dx = 0$

$$x^2 \cdot \frac{dy}{dx} + 2x = e^x$$

$$\frac{d^2 y}{dx^2} + \sin x = 0$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \sin xy$$

Ordinary Differential Equations (ODE)

A D.E containing a single independent variable

ie, the derivatives occurring in the eqs are ordinary derivatives.

* Order of a differential equation

It is the order of the highest ordered derivative occurring in it.

Degree of a Differential equation

It is the degree of the highest ordered derivative occurring in it, after the equation has been expressed in a form free from radicals and fractions

1) $\left(\frac{dy}{dx}\right)^2 + \frac{3d^2y}{dx^2} + y = \sin x$ as far as the derivatives are concerned.

order - 2

degree - 1

eg: 2) $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = \sin x$

order 2

degree 3

3) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = c^2 \cdot \frac{d^2y}{dx^2}$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left(c^2 \cdot \frac{d^2y}{dx^2} \right)^2$$

order - 2
degree - 2

* Homogeneous 2nd order linear ODE with constant coefficients

It can be written in the form,

$$y'' + ay' + by = 0$$

where a and b are constants.

OR

$$y'' = \frac{d^2y}{dx^2} \quad y' = \frac{dy}{dx}$$

In general, we can consider

n^{th} order linear ODE with constant coefficients

It can be written as,

$$\frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = 0$$

$$y^n + a_{n-1} y^{n-1} + \dots + a_1 y' + a_0 y = 0$$

where $a_0, a_1, a_2, \dots, a_{n-1}$ are constants.

* Solution of homogeneous linear ODE with constant coefficients.

Consider $y'' + ay' + by = 0$

To obtain the solution we first write the characteristic equation or auxiliary Equation.

It is given by $\lambda^2 + a\lambda + b = 0$ (by putting $y' = \lambda$, $y'' = \lambda^2$)

Let λ_1 and λ_2 be the roots of the auxiliary Equation. These are known as characteristic roots.

Case I : λ_1 and λ_2 are real & distinct.

Solution, $y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$

In general,

If λ_1, λ_2 etc... λ_n are the characteristic roots, all are real and distinct, solution is

~~$C_1 e^{\lambda_1 x}$~~

$$y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} + \dots C_n e^{\lambda_n x}$$

Case II

If λ_1 & λ_2 are real and equal

$$\lambda_1 = \lambda_2 = \lambda$$

Solution, $y = (C_1 + C_2 x) e^{\lambda x}$

(ii) If $\lambda_1 = \lambda_2 = \lambda_3 = \lambda$ and others different

$$(C_1 + C_2 x + C_3 x^2) e^{\lambda x} + C_4 e^{\lambda_4 x} + \dots C_n e^{\lambda_n x}$$

Case III

λ_1 and λ_2 are complex roots ($\pm \lambda i$ form)

Soln: $y = C_1 \cos \lambda x + C_2 \sin \lambda x$

If $\lambda = \alpha \pm i\beta$

Soln $y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x]$

$$\left. \begin{array}{l} \text{Q. } (\lambda^2 + 4)^2 = 0 \\ (C_1 + C_2 x) \cos 2x + (C_3 + C_4 x) \sin 2x \end{array} \right\} \text{mp}$$

Solve

$$y'' - y = 0$$

The characteristic eq is $\lambda^2 - 1 = 0$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

$$\lambda = +1, -1$$

Solution is $y = \underline{\underline{C_1 e^x + C_2 e^{-x}}}$

Solve,

$$y'' + y' - 2y = 0$$

The characteristic eq $\lambda^2 + \lambda - 2 = 0$

$$\lambda = 1, -2$$

Solution $y = \underline{\underline{C_1 e^x + C_2 e^{-2x}}}$

Solve,

$$y'' + 36y = 0$$

characteristic eq is $\lambda^2 + 36 = 0$

$$\lambda^2 = -36$$

$$\lambda = \pm 6i$$

$$\text{Solution } y = \underline{\underline{C_1 \cos 6x + C_2 \sin 6x}}$$

Solve

$$y'' + 4y' + 2.5y = 0$$

$$\lambda^2 + 4\lambda + 2.5 = 0$$

$$\lambda = \frac{-4 \pm \sqrt{6}}{2}$$

$$\underline{\underline{y = C_1 e^{\left(\frac{-4+\sqrt{6}}{2}\right)x} + C_2 e^{\left(\frac{-4-\sqrt{6}}{2}\right)x}}}$$

Solve

$$y^{iv} + 2y'' + y = 0$$

$$\lambda^4 + 2\lambda^2 + 1 = 0$$

$$\lambda^4 + \lambda^2 + \lambda^2 + 1 = 0$$

$$\lambda^2(\lambda^2 + 1) + (\lambda^2 + 1) = 0$$

$$(\lambda^2 + 1)(\lambda^2 + 1) = 0$$

$$(\lambda^2 + 1)^2 = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

$$\text{Solution } y = (C_1 + C_2 x) \cos x + \underline{\underline{(C_3 + C_4 x) \sin x}}$$

$$Q) \quad y''' + 9y' = 0$$

$$\lambda^3 + 9\lambda = 0$$

$$\lambda(\lambda^2 + 9) = 0$$

$$\lambda^2 = -9$$

$$\lambda = 0; \lambda = \pm 3i$$

$$y = (C_1 e^{0x} + C_2 \cos 3x + C_4 \sin 3x)$$

Case III

λ_1 & λ_2 are complex roots

$$8) y^4 + 16y'' = 0$$

$$\lambda^4 + 16\lambda^2 = 0 \quad (\text{characteristic eq})$$

$$\lambda^2[\lambda^2 + 16] = 0$$

$$\lambda^2 = 0 \quad \lambda^2 = -16$$

$$\lambda = 0, 0 \quad \lambda = \pm 4i$$

Solution,

$$\underline{y = (C_1 + C_2 x) e^{0x} + C_3 \cos 4x + C_4 \sin 4x}$$

$$9) \cancel{D^2} [D^3 - D^2 - D + 1] y = 0$$

$$ans) \lambda^3 - \lambda^2 - \lambda + 1 = 0 \quad (\text{characteristic eq})$$

$$\lambda = -1, 1, 1$$

$$\text{Solution, } y = \underline{C_1 e^{-x} + (C_2 + C_3 x) e^x}$$

$$Q) (D^4 + 10D^2 + 9)y = 0$$

$$Q) (D^5 + 20D^3 + D)y = 0$$

$$Q) y'' + 9y = 0$$

$$Q) y'' - 3y' - 4y = 0$$

$$Q_1) y'' + 3y' + y = 0 \quad 4y = 0$$

$$\lambda^2 + 3\lambda + 1 = 0$$

$$Q_2) y'' + 2\pi y' + \pi^2 y = 0$$

$$Q_1) (D^4 + 10D^2 + 9)y = 0$$

Auxiliary equation is

$$\lambda^4 + 10\lambda^2 + 9 = 0$$

$$\lambda^4 + 10\lambda^2 + 9 = 0$$

$$(\lambda^2)^2 + 10\lambda^2 + 9 = 0$$

$$(\lambda^2)^2 + 10\lambda^2 + 9 = 0$$

$$\lambda^2(\lambda^2 + 5)^2 - 16 = 0$$

$$\lambda = \pm 4, \pm 5i$$

Solution is

$$y = C_1 e^{4x} + C_2 e^{-4x} + C_3 \cos 5x + C_4 \sin 5x$$

$$Q_2) (D^5 + 2D^3 + D)y = 0$$

Auxiliary eq is

$$\lambda^5 + 2\lambda^3 + \lambda = 0$$

$$\lambda(\lambda^4 + 2\lambda^2 + 1) = 0$$

$$\lambda = 0 \quad \& \quad (\lambda^2 + 1)^2 = 0$$

$$\lambda = 0 \quad \& \quad (\lambda^2 + 1) = 0$$

$$\lambda = 0 \quad \& \quad \lambda = \pm i, \pm i,$$

Soln : is,

$$y = C_1 e^{0x} + (C_2 + C_3 x) \cos x + \underline{(C_4 + C_5 x) \sin x}$$

Q) $y'' + 9y = 0$

(Auxiliary eq) = is $\lambda^2 + 9 = 0$

$$\lambda = \pm 3i$$

Solution is

$$\underline{y = C_1 \cos 3x + C_2 \sin 3x}$$

Q) $y'' - 3y' - 4y = 0$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$\lambda^2 - 4\lambda + \lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0$$

$$\lambda = -1, 4$$

$$\text{Soln is } \underline{y = c_1 e^{-x} + c_2 e^{4x}}$$

$$Q) \quad y'' + 2\pi y' + \pi^2 y = 0$$

Auxiliary eq is

$$\lambda^2 + 2\pi\lambda + \pi^2 = 0$$

$$(\lambda + \pi)^2 = 0$$

$$\lambda + \pi = 0$$

$$\lambda = -\pi, -\pi$$

$$\text{Soln is } \underline{y = (c_1 + c_2 x) e^{-\pi x}}$$

Solution of Initial Value Problem (IVP)

$$Q) \quad y'' + 9y = 0 \quad \text{where } y(0) = 0.2 \quad y'(0) = -1.5$$

Solve I.V.P

Auxiliary Eqn $\lambda^2 + 9 = 0$

$$\lambda^2 = -9$$

$$\lambda = \pm 3i$$

$$y = C_1 \cos 3x + C_2 \sin 3x$$

$$\underline{y(x) = C_1 \cos 3x + C_2 \sin 3x} \longrightarrow \textcircled{1}$$

$$\cancel{y(0) =}$$

$$y'(x) = C_1 - \sin 3x \cdot 3 + C_2 \cos 3x \cdot 3$$

$$= -3C_1 \sin 3x + 3C_2 \cos 3x \longrightarrow \textcircled{2}$$

Given that $y(0) = 0.2$ $y'(0) = -1.5$

$$\cancel{y(0)}$$

$$y(0) = C_1 \cos(0) + C_2 \sin(0) = 0.2$$

$$= C_1 + 0 = 0.2$$

$$\underline{C_1 = 0.2}$$

$$\cancel{y'(-1.5)}$$

$$y'(0) = -3C_1 \sin(0) + 3C_2 \cos(0) = -1.5$$

$$= 3C_2 = -1.5$$

$$C_2 = \frac{-1.5}{3} = \underline{\underline{-0.5}}$$

$\therefore$ Soln of I.V.P

$$\underline{\underline{y = 0.2 \cos 3x - 0.5 \sin 3x}}$$

Solve the I.V.P

$$y'' - 3y' - 4y = 0 \text{ where } y(0) = 2, y'(0) = 1$$

Auxiliary Eqn

$$\lambda^2 - 3\lambda - 4 = 0$$

$$\lambda = 4, -1$$

$$\text{Soln } y = C_1 e^{4x} + C_2 e^{-x}$$

$$y(x) = C_1 e^{4x} + C_2 e^{-x} \longrightarrow \textcircled{1}$$

$$y'(x) = 4C_1 e^{4x} - C_2 e^{-x} \longrightarrow \textcircled{2}$$

$$y(0) \Rightarrow C_1 e^0 + C_2 e^0 = 2$$

$$\Rightarrow C_1 + C_2 = 2$$

$$y'(0) = 4C_1 e^0 - C_2 e^0 = 1$$

$$4C_1 - C_2 = 1 \longrightarrow \textcircled{3}$$

$$C_1 + C_2 = 2 \longrightarrow \textcircled{4}$$

$$\underline{\underline{C_1 = 3/5, C_2 = 7/5}}$$

Solution of I.V.P

$$\underline{\underline{y = \frac{3}{5} e^{4x} + \frac{7}{5} e^{-x}}}$$

Solve the I.V.P

$$y'' - y = 0 \quad \text{given that } y(0) = 2 \quad y'(0) = -2$$

$$\lambda^2 - 1 = 0$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

$$\text{Soln } y = C_1 e^x + C_2 e^{-x}$$

$$y(x) = C_1 e^x + C_2 e^{-x} \longrightarrow (1)$$

$$y'(x) = C_1 e^x - C_2 e^{-x} \longrightarrow (2)$$

$$y(0) = C_1 e^0 + C_2 e^0 = 2$$

$$= C_1 + C_2 = 2 \longrightarrow (3)$$

$$y'(0) = C_1 e^0 - C_2 e^0 = -2$$

$$= C_1 - C_2 = -2 \longrightarrow (4)$$

$$\underline{\underline{C_1 = 0}}$$

$$\underline{\underline{C_2 = 2}}$$

$$\text{Soln of I.V.P } y = 2e^{-x}$$

Solve,

$$y^{iv} - 9y'' - 400y = 0$$

$$y(0) = 0, y'(0) = 0, y''(0) = 41, y'''(0) = 0$$

Auxiliary Eqn

$$\lambda^4 - 9\lambda^2 - 400 = 0$$

$$\lambda(\lambda^3 - 9\lambda)$$

$$\text{Put } \lambda^2 = t$$

$$t^2 - 9t - 400 = 0$$

$$\rightarrow t = 25, -16$$

$$\rightarrow (t - 25)(t + 16) = 0$$

$$(\lambda^2 - 25)(\lambda^2 + 16) = 0$$

$$\lambda^2 = 25 \quad \lambda^2 = -16$$

$$\lambda = \pm 5$$

$$\lambda = \pm 4i$$

$$\text{Soln, } y = (C_1 e^{5x} + C_2 e^{-5x}) + (C_3 \cos 4x + C_4 \sin 4x)$$

$$y(x) = (C_1 e^{5x} + C_2 e^{-5x}) + (C_3 \cos 4x + C_4 \sin 4x)$$

→ ①

$$y'(x) = 5C_1 e^{5x} - 5C_2 e^{-5x} - 4C_3 \sin 4x + 4C_4 \cos 4x \rightarrow (2)$$

$$y''(x) = 25C_1 e^{5x} + 25C_2 e^{-5x} - 16C_3 \cos 4x - 16C_4 \sin 4x \rightarrow (3)$$

$$y'''(x) = 125C_1 e^{5x} - 125C_2 e^{-5x} + 64C_3 \sin 4x - 64C_4 \cos 4x \rightarrow (4)$$

Put $x=0$ in eq (1) & (2), (3), (4)

$$y(0) = C_1 e^0 + C_2 e^0 + C_3 \cos 0 + C_4 \sin 0$$

$$\Rightarrow C_1 + C_2 + C_3 = 0 \rightarrow (5)$$

$$y'(0) = 5C_1 - 5C_2 + 4C_4 = 0 \rightarrow (6)$$

$$y''(0) = 25C_1 + 25C_2 - 16C_3 = 41 \rightarrow (7)$$

$$y'''(0) = 125C_1 - 125C_2 - 64C_4 = 0 \rightarrow (8)$$

$$C_1 = -C_3 - C_2 \text{ from eq (5)}$$

$$5(-C_3 - C_2) - 5C_2 + 4C_4 = 0$$

$$-5C_3 - 5C_2 - 5C_2 + 4C_4 = 0$$

$$-5C_3$$

5 & 7 (Eq) Comparing

$$25 \times (5) - (7) \Rightarrow$$

$$0 = 25C_1 + 25C_2 + 25C_3$$

$$41 = 25C_1 + 25C_2 - 16C_3$$

$$\underline{-41 = 41C_3}$$

$$25 - (-16)$$

$$\underline{\underline{C_3 = -1}}$$

(6) & (8) Eq Comparing

$$25 \times (6) - (8) \Rightarrow$$

$$125C_1 - 125C_2 + 100C_4 = 0$$

$$\underline{-125C_1 - 125C_2 - 64C_4 = 0}$$

$$C_4 = 0$$

$$C_1 + C_2 + C_3 = 0$$

In Eq 5 & 6

$$0 = C_1 + C_2 - 1$$

$$0 = 5C_1 - 5C_2 - 4(0)$$

$$C_1 + C_2 = 1$$

$$5C_1 - 5C_2 = 0$$

$$\underline{\underline{C_1 = \frac{1}{2} ; C_2 = \frac{1}{2}}}$$

Soln,

$$\underline{\underline{y(x) = \left(\frac{1}{2} e^{5x} + \frac{1}{2} e^{-5x}\right) + (-\cos 4x) + 0 \sin 4x}}$$

Q Solve

$$y'' + y' + 3.25(y) = 0$$

$$\lambda^2 + \lambda + 3.25 = 0 \quad (\text{Characteristic eq})$$

$$\lambda = \frac{-1 \pm \sqrt{1 - 13}}{2} = \frac{-1 \pm \sqrt{-12}}{2} = \frac{-1 \pm 2\sqrt{3}i}{2}$$

$$\text{Soln. } \underline{\underline{y(x) = e^{-1/2x} [C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x]}}$$

$$\text{Solve } y'' + .54 y' + (0.0729 + \pi)y = 0$$

$$\lambda^2 + .54\lambda + (0.0729 + \pi) = 0$$

$$\lambda = \frac{-.27}{100} \pm 1.7724 i \times$$

OR

$$-0.27 \pm \sqrt{\pi} i$$

$$y = e^{-0.27x} (C_1 \cos \sqrt{\pi}x + C_2 \sin \sqrt{\pi}x)$$

Solve

$$100y'' + 240y' + (196\pi^2 + 144)y = 0$$

$$100\lambda^2 + 240\lambda + 196\pi^2 + 144 = 0$$

$$\lambda = \frac{-6}{5} \pm \frac{7}{5}\pi i$$

$$y = e^{-6/5x} (C_1 \cos 7/5 \pi x + C_2 \sin 7/5 \pi x)$$

Q) Solve the IVP,

$$y'' + 0.4y' + 9.04y = 0 \text{ where } y(0) = 0$$

$$y'(0) = 3$$

$$\lambda^2 + 0.4\lambda + 9.04 = 0$$

$$\lambda = -\frac{1}{5} \pm 3i$$

General solution

$$y(x) = e^{-1/5x} (C_1 \cos 3x + C_2 \sin 3x)$$

$$y(0) = e^0 [C_1 \cos(0) + C_2 \sin(0)] = 0$$

$$\Rightarrow \underline{C_1 = 0}$$

$$y'(x) = e^{-1/5x} [-3C_1 \sin 3x + 3C_2 \cos 3x] +$$
$$-\frac{1}{5} e^{-1/5x} [C_1 \cos 3x + C_2 \sin 3x]$$

$$y'(0) = 3$$

$$e^0 [-3C_1 \sin(0) + 3C_2 \cos(0)] + \\ - \frac{1}{5} e^0 [C_1 \cos(0) + C_2 \sin(0)]$$

$$3C_2 + -\frac{1}{5}[C_1] = 3$$

$$3C_2 - \frac{C_1}{5} = 3$$

$$3C_2 = 3$$

$$\underline{\underline{C_2 = 1}}$$

$$\underline{\underline{y(x) = e^{-1/5x} \cdot \sin 3x}} \quad \text{(particular solution)}$$

General solution & Particular soln. of a
Second order linear homogeneous ODE.

Let $y'' + P(x)y' + Q(x)y = 0$ be the second order linear homogeneous ODE $\longrightarrow \textcircled{1}$

A general solution of 1 is of the form, $y(x)$

$$y(x) = C_1 y_1 + C_2 y_2 \quad \text{where } y_1 \text{ \& } y_2 \longrightarrow \textcircled{2}$$

are solutions of eq ① that are not multiples of each other and C_1 & C_2 are arbitrary constants.

Particular solution of one is obtained if we assign specific values to C_1 & C_2 in equation no. ②

Super position principle for Second order linear homogeneous ODE.

- * The super position principle says that we can obtain further solutions from given ones by adding them or by multiplying them with any constant. In particular for such an equation ~~some~~ sums and constant multiples of solutions are again solutions.
- * i.e., if y_1 & y_2 are solutions to a homogeneous 2nd order linear ODE then
$$y = C_1 y_1 + C_2 y_2$$
is also a solution of the same.

* Super position principle is true only for homogeneous ODE.

Q) Does the function $y = C_1 \cos 2x + C_2 \sin 2x$ form a solution of $y'' + 4y = 0$.
Is it the general soln. Justify your answer.

ans) Consider,

$$y_1 = \cos 2x$$

$$y_1' = -2 \sin 2x$$

$$y_1'' = -4 \cos 2x$$

$$y_1'' + 4y_1 = 0$$

$$-4 \cos 2x + 4 \cos 2x = 0$$

So, $y_1 = \cos 2x$ is a soln of $y'' + 4y = 0$

Consider $y_2 = \sin 2x$

$$y_2' = 2 \cos 2x$$

$$y_2'' = -4 \sin 2x$$

$$-4 \sin 2x + 4 \sin 2x = 0$$

So $y_2 = \sin 2x$ is also a soln.

Consider, y_1 & y_2 are independent linearly

$$y = C_1 \cos 2x + C_2 \sin 2x$$

$$\frac{y_1}{y_2} = \frac{\cos 2x}{\sin 2x} = \cot 2x \neq \text{const.}$$

$$y' = -2C_1 \sin 2x + 2C_2 \cos 2x$$

$$y'' = -4C_1 \cos 2x - 4C_2 \sin 2x$$

$$\begin{aligned} y'' + 4y &= -4C_1 \cos 2x - 4C_2 \sin 2x + \\ &\quad 4[C_1 \cos 2x + C_2 \sin 2x] \\ &= -4C_1 \cos 2x - 4C_2 \sin 2x + 4C_1 \cos 2x \\ &\quad + 4C_2 \sin 2x \\ &= \underline{\underline{0}} \end{aligned}$$

By superposition principle

y is a ~~soln~~ general soln becoz there are 2 arbitrary constants.

$$y = C$$

Q) Verify by substitution that $y_1 = e^x$, $y_2 = e^{-x}$ are solutions of the ODE $y'' - y = 0$

Solve the initial value problem $y'' - y = 0$
when $y(0) = 6$ $y'(0) = -2$

$$y_1 = e^x$$

$$y_2 = e^{-x}$$

$$y_1' = e^x$$

$$y_2' = -e^{-x}$$

$$y_1'' = e^x$$

$$y_2'' = e^{-x}$$

$$y_2'' - y_2 = e^{-x} - e^{-x} = 0$$

$$y_1'' - y_1 = e^x - e^x = 0$$

$\therefore y_1$ & y_2 are the soln of eq.

$$\frac{y_1}{y_2} = \frac{e^x}{e^{-x}} = e^x \cdot e^x = e^{2x} \neq \text{const.}$$

$\therefore y_1$ & y_2 are independent of each other.

$\therefore$ By superposition principle,

general soln is

$$\underline{y = C_1 e^x + C_2 e^{-x}}$$

$$y'(x) = C_1 e^x - C_2 e^{-x}$$

$$y(0) = C_1 e^0 + C_2 e^0 = 6$$

$$= C_1 + C_2 = 6 \longrightarrow \textcircled{1}$$

$$y'(0) = C_1 - C_2 = -2 \longrightarrow \textcircled{2}$$

$$C_1 = -2 + C_2 \longrightarrow (a)$$

Sub. (a) in $\textcircled{1}$

$$-2 + C_2 + C_2 = 6$$

$$\frac{C_2 = 4}{C_1 = 2}$$

particular soln is

$$y = \underline{2e^x + 4e^{-x}}$$

Euler - Cauchy Equation

* Euler Cauchy equations are ODE of the form

$$x^2 y'' + a x y' + b y = 0$$

Auxiliary Eq. is given by

$$\lambda^2 + (a-1)\lambda + b = 0$$

$\lambda = \lambda_1, \lambda_2$ are the characteristic roots

Ch. roots	Nature	General solution
1) $\lambda = \lambda_1, \lambda_2$	real & distinct	$y(x) = C_1 e^{x_1} + C_2 e^{x_2}$
2) $\lambda_1 = \lambda_2 = \lambda$	real & same	$y(x) = C_1 + C_2 \ln(x) \cdot x^\lambda$ $(C_1 + C_2 \ln(x)) x^\lambda$
3) $\lambda = \lambda \pm i\beta$	complex conjugate pair	$y(x) = x^\lambda [C_1 \cos \beta \ln x + C_2 \sin \beta \ln x]$

Note: -

x is replaced by $\ln(x)$

Solve,

$$x^2 y'' - 2xy' = 0$$

1) $x^2 y'' - 2y = 0$

2) $5x^2 y'' + 23xy' + 16 - 2y = 0$

3) $xy'' + 4y' = 0$

4) $x^2 y'' + 1.5xy' - 0.5y = 0$

5) $x^2 y'' - 3xy' + 9y = 0$

6) $x^2 y'' + 0.6xy' + 16.04y = 0$

7) $4x^2 y'' + 5y = 0$

8) $x^2 y'' + 0.7xy - (0.1)y = 0$

9) ~~$(x^2 D^2 - xD - 5I)y = 0$~~
 $(x^2 D^2 - xD - 5I)y = 0$

1) $x^2 y'' - 2y = 0$

$$x^2 y'' + axy' + by = 0 \quad (\text{Euler-Cauchy eq})$$

$$a = 0, \quad b = -2$$

$$\lambda^2 - (0-1)\lambda + (-2) = 0$$

$$\lambda^2 + \lambda - 2 = 0$$

$$\left[\begin{array}{l} \lambda^2 - (a-1)\lambda + b = 0 \\ \text{characteristic eq} \end{array} \right]$$

$$\underline{\lambda = -1, +2}$$

General soln is

$$\underline{C_1 x^2 + C_2 x^{-1}} \text{ or } \underline{C_2 x^2 + \frac{C_2}{x}}$$

$$2) \quad 5x^2y'' + 23xy' + 16 \cdot 2y = 0$$

Divide throughout by 5

$$x^2y'' + \frac{23}{5}xy' + \frac{16 \cdot 2}{5}y = 0 \quad (\text{st. form of Euler cauchy eq})$$

$$\cancel{x^2}y''$$

$$a = \frac{23}{5} \quad b = \frac{16 \cdot 2}{5}$$

$$\lambda^2 + (a-1)\lambda + b = 0$$

$$\lambda^2 + \left(\frac{23}{5} - 1\right)\lambda + \frac{16 \cdot 2}{5} = 0$$

$$\lambda^2 + \frac{18}{5}\lambda + \frac{16 \cdot 2}{5} = 0$$

$$\lambda = \underline{\underline{-\frac{9}{5}, -\frac{9}{5}}}$$

General soln

$$\underline{\underline{y(x) = (C_1 + C_2 \ln x) x^{-9/5}}}$$

Solve the IVP

3) Solve $xy'' + 4y' = 0$

Multiply by x .

$$x^2 y'' + 4xy' = 0$$

$$a = 4, \quad b = 0$$

Auxillary eq: $\lambda^2 + (a-1)\lambda + b = 0$

$$\lambda^2 + 3\lambda + 0 = 0$$

$$\lambda^2 = -3\lambda$$

$$\lambda = 0, -3$$

General soln = $y(x) = C_1 x^0 + C_2 x^{-3}$

$$\underline{\underline{y(x) = C_1 + C_2/x^3}}$$

4) $x^2 y'' + 1.5xy' - 0.5y = 0$

$$a = 1.5 \quad b = -0.5$$

$$\lambda^2 + (a-1)\lambda + b = 0$$

$$\lambda^2 - (1.5-1)\lambda - 0.5 = 0$$

$$\lambda = 1/2, -1$$

General soln

$$y(x) = C_1 x^{1/2} + C_2 x^{-1} \\ = \underline{\underline{C_1 \sqrt{x} + \frac{C_2}{x}}}$$

$$5) \quad x^2 y'' - 5xy' + 9y = 0$$

$$a = -5, \quad b = 9$$

$$\lambda^2 - (a-1)\lambda + b = 0 \quad \lambda^2 - 6\lambda + 9 = 0$$

$$\lambda^2 - (-5-1)\lambda + 9 = 0 \quad \lambda = 3, 3$$

$$\lambda^2 - (-6)\lambda + 9 = 0$$

$$\lambda^2 + 6\lambda + 9 = 0$$

$$y(x) = \underline{\underline{C_1 x^3 + C_2 x^3}} \\ \underline{\underline{(C_1 + C_2 \ln x) x^3}}$$

$$6) \quad x^2 y'' + 0.6xy' + 16.04y = 0$$

$$a = 0.6 \quad b = 16.04$$

$$\lambda^2 + (a-1)\lambda + b = 0$$

$$\lambda^2 - 0.4\lambda + 16.04 = 0$$

$$\lambda = \frac{1}{5} \pm 4i$$

$$y(x) = \underline{\underline{x^{1/5} (C_1 \cos(4 \ln x) + C_2 \sin(4 \ln x))}}}$$

$$7) 4x^2 y'' + 5y = 0$$

$$x^2 y'' + \frac{5}{4} y = 0$$

$$\lambda^2 + \frac{5}{4} = 0$$

$$\lambda^2 = -\frac{5}{4}$$

$$\lambda = \pm \sqrt{\frac{5}{4}} i$$

$$\lambda^2 - \lambda + \frac{5}{4} = 0$$

$$\lambda = \frac{1}{2} \pm i$$

$$\text{Soln: } (y(x) = x^{1/2} (C_1 \cos \ln x + C_2 \sin \ln x))$$

$$\underline{\underline{y(x) = C_1 \cos \sqrt{5}/2 (\ln x) + C_2 \sin \sqrt{5}/2 (\ln x)}}$$

$$8) x^2 y'' + 0.7 y - (0.1) y = 0$$

$$\lambda^2 + (0.7 - 1) \lambda + (0.1) = 0$$

$$\lambda^2 - 0.3 \lambda - 0.1 = 0$$

$$\lambda = \frac{1}{2}, -\frac{1}{5}$$

$$y(x) = C_1 x^{1/2} + C_2 x^{-1/5} = \underline{\underline{C_1 \sqrt{x} + \frac{C_2}{x^{1/5}}}}$$

$$9) (x^2 D - x D - 5I) y = 0$$

$$x^2 y'' - x y' - 5y = 0$$

$$\lambda^2 + (-1-1) \lambda + (-5) = 0$$

$$\lambda^2 - 5 = 0$$

$$\lambda^2 = 5$$

$$\lambda^2 - 2\lambda - 5 = 0$$

$$\lambda = 1 + \sqrt{6}, 1 - \sqrt{6}$$

$$\lambda = \pm \sqrt{5}$$

$$y(x) = \underline{C_1 x^{\sqrt{5}} + C_2 x^{-\sqrt{5}}}$$

$$y = \underline{x [C_1 \cos(\sqrt{6} \ln x) + C_2 \sin(\sqrt{6} \ln x)]}$$

Q) Solve the initial value problem,

$$x^2 y'' - 4xy' + 6y = 0 \quad y(1) = 0.4, \quad y'(1) = 0$$

$$x^2 y'' - 4xy' + 6y = 0$$

$$a = -4, \quad b = 6$$

$$\lambda^2 + (a-1)\lambda + b = 0$$

$$\lambda^2 + (-4-1)\lambda + 6 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda = 2, 3$$

$$y(x) = \underline{C_1 x^2 + C_2 x^3}$$

$$y'(x) = 2C_1 x + 3C_2 x$$

$$y(1) = \cancel{C_1} + \cancel{C_2} \quad C_1(1)^3 + C_2(1)^2 = 0.4$$

$$y'(1) = 2C_1(1) + 3(C_2)(1) = 0$$

$$C_1 + C_2 = 0.4 \longrightarrow \textcircled{1}$$

$$2C_1 + 3C_2 = 0 \longrightarrow \textcircled{2}$$

$$\textcircled{1} \times 2 \Rightarrow 2C_1 + 2C_2 = 0.8 \quad -$$

$$2C_1 + 3C_2 = 0$$

$$-1C_2 = 0.8$$

$$\underline{\underline{C_2 = -0.8}}$$

$$C_1 - 0.8 = 0.4$$

$$C_1 = 0.4 + 0.8 = \underline{\underline{1.2}}$$

Particular soln $\Rightarrow$

$$\cancel{y(x) = 1.2x + 0}$$

$$\underline{\underline{y(x) = 1.2x^2 - 0.8x^3}}$$

Solve the IVP,

$$(9x^2 D^2 + 3xD + I)y = 0 \quad y(1) = 1, \quad y'(1) = 0$$

$$9x^2 y'' + 3xy' + y = 0$$

$$x^2 y'' + \frac{1}{3} x y' + \frac{1}{9} y = 0 \quad a = \frac{1}{3}, b = \frac{1}{9}$$

$$\lambda^2 + \left(\frac{1}{3} - 1\right)\lambda + \frac{1}{9} = 0$$

$$\lambda^2 + \left(-\frac{2}{3}\right)\lambda + \frac{1}{9} = 0$$

$$\lambda = \frac{1}{3}, \frac{1}{3}$$

general soln =

$$y(x) = \underline{(C_1 + C_2 \ln x) x^{1/3}} \rightarrow \textcircled{1}$$

$$y'(x) = (C_1 + C_2 \ln x) \frac{1}{3} x^{-2/3} + (0 + \frac{C_2}{x}) x^{1/3} \rightarrow \textcircled{2}$$

$$y(1) = (C_1 + C_2 \ln(1))^{1/3}$$

$$\Rightarrow C_1 = 1$$

$$y'(1) = (C_1 + C_2 \ln(1)) \frac{1}{3} 1^{-2/3} + (0 + \frac{C_2}{1}) 1^{1/3}$$

$$= 1 + \frac{C_2}{3}$$

$$C_2 = -1/3$$

$$\underline{C_2 = -1/3}$$

$$\underline{y(x) = (1 - \frac{1}{3} \ln x) x^{1/3}}$$

Q) Solve IVP

$$x^2 y'' + x y' + 9y = 0 \quad y(1) = 0$$

$$y'(1) = 2.5$$

$$a = 1 \quad b = 9$$

$$\lambda^2 + (a-1)\lambda + 9 = 0$$

$$\lambda^2 + 9 = 0$$

$$\lambda^2 = -9$$

$$\lambda = \pm 3i$$

general soln

$$y(x) = C_1 \cos(3 \ln x) + C_2 \sin(3 \ln x) \rightarrow \textcircled{1}$$

$$y'(x) = -3C_1 \sin \ln x \cdot \frac{1}{x} + 3C_2 \cos(\ln x) \cdot \frac{1}{x}$$

$$y(1) = C_1 \cos(3 \ln 1) + C_2 \sin(3 \ln 1)$$

$$\Rightarrow \underline{C_1 = 0}$$

$$y'(1) = -3C_1 \sin(\ln 1) \cdot \frac{1}{1} + 3C_2 \cos(\ln 1) \cdot \frac{1}{1} = 2.5$$

$$\Rightarrow \underline{0} + 3C_2 \cdot 1 = 2.5$$

$$C_2 = 2.5/3 = \underline{\underline{5/6}}$$

$$\underline{\underline{y(x) = 5/6 \sin(3 \ln x)}}$$

Non Homogeneous O.D.E (Right Hand Side $\neq 0$)

Consider the second order the non homogeneous linear ODE

$$y'' + p(x)y' + q(x)y = r(x) \rightarrow (1)$$

where $r(x) \neq 0$

A general soln of eq (1) is of the form,

$$y(x) = y_h(x) + y_p(x)$$

where $y_h(x)$ is the general soln of homogeneous linear ODE containing two arbitrary constant and $y_p(x)$ is a any soln of eq (1) containing no arbitrary constants.

In order to find out $y_p(x)$ there ^{are} ~~are~~ ~~two~~ two methods.

- * Method 1 : Variation of parameters
- * Method 2 : Method of undetermined coefficients

Variation of Parameters: The method

Consider $y'' + p(x)y' + q(x)y = r(x)$
general soln,

$$\cancel{y(x) = y_h(x) + y_p(x)}$$

$$\cancel{y(x) = y_h(x) + y_p(x)} \Rightarrow \cancel{y_h(x) = C_1 y_1 + C_2 y_2}$$

$$\cancel{y_p(x) = U y_1 + V y_2}$$

$$y(x) = y_h(x) + y_p(x)$$

$$y_h(x) = C_1 y_1 + C_2 y_2$$

$$y_p(x) = U y_1 + V y_2$$

$$\text{Let } y_h(x) = C_1 y_1 + C_2 y_2,$$

$$\text{where } U = \int \frac{y_2 \text{ RHS}}{W} dx$$

$$V = \int \frac{y_1 \text{ RHS}}{W} dx$$

$$\text{where Wronskian} = W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Solve: $y'' + y = \sec x$

by method of variation of parameters

$$y(x) = y_h(x) + y_p(x)$$

To find $y_h(x) \Rightarrow y'' + y = 0$

auxiliary eq: $\lambda^2 + 1 = 0$
 $\lambda = \pm i$

h-homogeneous: $y_h(x) = C_1 \cos x + C_2 \sin x$ $[C_1 y_1 + C_2 y_2]$

$y_1 = \cos x$ $y_2 = \sin x$

$$y_p(x) = u y_1 + v y_2$$

$$u = - \int \frac{y_2 \cdot \text{RHS}}{W} dx$$

$$W = \begin{vmatrix} y_1 & y_1' \\ y_2 & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{vmatrix}$$

$$= \cos^2 x + \sin^2 x$$

$$= 1$$

$$u = - \int \frac{\sin x \cdot \sin x}{1}$$

$$u = - \int \frac{\sin x \cdot \sec x}{1}$$

$$= - \int \frac{\sin x}{\cos x}$$

$$= \bullet \log |\cos x|$$

$$\begin{cases} \frac{\int f'(x) dx}{f(x)} \\ = \log |f(x)| + c \end{cases}$$

$$\text{Stam} = \log |\sec x| \text{ or } -\log |\cos x|$$

$$V = \int \frac{y_1 \cdot R.H.S}{W} dx$$

$$= \int \frac{\cos x \cdot \sec x}{1} dx$$

$$= \int \frac{\cos x}{\cos x} dx$$

$$= \int dx$$

$$= \underline{\underline{x}}$$

$$y_p(x) = \log |\cos x| \cos x + x \cdot \sin x \rightarrow (2)$$

General solution,

$$u(x) = y_h(x) + y_p(x)$$

$$= \underline{C_1 \cos x + C_2 \sin x + C \cos x \log |\cos x| + x \sin x}$$

Q, Solve by method of variation of parameters

$$y'' + 4y = \cos 2x$$

$$y(x) = y_h(x) + y_p(x)$$

To find, $y_h(x) = y'' + 4y = 0$

$$\text{auxiliary eq} = \lambda^2 + 4 = 0$$

$$\lambda^2 = -4$$

$$\lambda = \pm 2i$$

$$\underline{y_h(x) = C_1 \cos 2x + C_2 \sin 2x}$$

$$y_1 = \cos 2x$$

$$y_2 = \sin 2x$$

$$y_p(x) = u y_1 + v y_2$$

$$u = - \int \frac{y_2 \cdot \text{RHS}}{W} dx$$

$$W = \begin{vmatrix} y_1 & y_1' \\ y_2 & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & -2 \sin 2x \\ \sin 2x & 2 \cos 2x \end{vmatrix}$$

$$= 2 \cos^2 2x + 2 \sin^2 2x$$

$$= 2 (\cos^2 2x + \sin^2 2x) = \underline{2}$$

$$u = - \int \frac{\sin 2x \cdot \cos 2x}{2} dx$$

$$= - \int \frac{\sin 4x}{4} dx$$

$$= -\frac{1}{4} \int \sin 4x = \frac{\cos 4x}{4} \left(-\frac{1}{4} \right) = \underline{\underline{\frac{\cos 4x}{16}}}$$

(smaller
2 sin 2 cos 2)

$$\begin{aligned} \sin 2x \cdot \cos 2x \\ = \frac{\sin 4x}{2} \end{aligned}$$

$$v = \int \frac{y \cdot RHS}{w} dx$$

$$= \int \frac{\cos 2x \cdot \cos 2x}{2} dx$$

$$= \int \frac{\cos^2 2x}{2} dx$$

$$= \int \frac{1 + \cos 4x}{2 \times 2} dx$$

$$\left(\cos^2 x = \frac{1 + \cos 2x}{2} \right)$$

$$= \frac{1}{4} \left[\frac{\sin 4x}{4} + x \right] = \frac{\sin 4x}{16} + \frac{x}{4}$$

$$y_p(x) = \frac{\cos 4x}{16} \cdot \cos 2x + \frac{1}{16} (4x + \sin 4x) \sin 2x$$

$$= \frac{\cos 4x \cos 2x + \sin 4x \sin 2x}{16} + \frac{4x}{16} \sin 2x$$

$$= \frac{\cos(4x - 2x)}{16} + \frac{x \sin 2x}{4} \left[\begin{array}{l} \cos(A - B) \\ = \cos A \cos B + \\ \sin A \sin B \end{array} \right]$$

General soln:

$$y(x) = y_h(x) + y_p(x)$$

$$= (C_1 \cos 2x + C_2 \sin 2x) + \frac{\cos 2x}{16} + \frac{x \sin 2x}{4}$$

Solve

$$Q) (D^2 + 6D + 9I)y = \frac{16e^{-3x}}{x^2 + 1}$$

[M.V.P is applicable only when R.H.S is a complex function]
becomes complicated.

$$y'' + 6y' + 9 = \frac{16e^{-3x}}{x^2 + 1}$$

$$y'' + 6y' + 9 = 0$$

$$\lambda^2 + 6\lambda + 9 = 0 \quad [\text{characteristic eq}]$$

$$\lambda = -3, -3$$

$$y_h(x) = (C_1 + C_2 x) e^{-3x}$$

$$= \underline{C_1 e^{-3x} + C_2 e^{-3x} x}$$

$$y_1 = e^{-3x} \quad ; \quad y_2 = x e^{-3x}$$

$$w = \begin{vmatrix} y_1 & y_1' \\ y_2 & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} e^{-3x} & -3e^{-3x} \\ x e^{-3x} & x(-3e^{-3x}) + e^{-3x} \end{vmatrix}$$

$$= -3x e^{-3x} \cdot e^{-3x} + e^{-3x} e^{-3x} + 3x e^{-3x} e^{-3x}$$

$$= \underline{e^{-6x}}$$

$$y_p(x) = u y_1 + v y_2$$

$$u = - \int \frac{y_2 \cdot \text{RHS}}{w} dx$$

$$= - \int \frac{x e^{-3x} \cdot \frac{16 e^{-3x}}{x^2+1}}{e^{-6x}} dx$$

$$= -16 \int \frac{x e^{-6x}}{e^{-6x} \cdot x^2+1}$$

$$= -16 \int \frac{x}{x^2+1} dx$$

$$= -16 \cdot \frac{1}{2} \log |x^2+1|$$

$$= \underline{\underline{-8 \log |x^2+1|}}$$

$$v = \int \frac{y_1 \cdot R.H.S}{w} dx$$

$$= \int \frac{e^{-3x} \cdot \frac{16 e^{-3x}}{x^2+1}}{e^{-6x}} dx$$

$$= 16 \int \frac{1}{x^2+1} dx$$

$$= \underline{\underline{16 \tan^{-1} x}}$$

$$y_p(x) = (-8 \log|x^2+1|)e^{-3x} + (16 \tan^{-1}x)x e^{-3x}$$

General soln:

$$y(x) = C_1 + C_2$$

$$y(x) = C_1 e^{-3x} + C_2 e^{-3x} x +$$

$$(-8 \log|x^2+1|)e^{-3x} + (16 \tan^{-1}x)x e^{-3x}$$

$$Q4) y'' + 9y = \operatorname{cosec} 3x$$

$$Q5) (D^2 + 2D + 2I)y = 4e^{-x} \sec 3x$$

$$Q6) (D^2 - 2D + I)y = 35 x^{3/2} e^x$$

$$Q4) y'' + 9y = \operatorname{cosec} 3x$$

$$y'' + 9y = 0$$

$$\lambda^2 + 9 = 0$$

$$\lambda^2 = -9$$

$$\lambda = \pm 3i$$

$$y_h(x) = \underline{C_1 \cos 3x + C_2 \sin 3x}$$

$$y_1 = \cos 3x \quad y_2 = \sin 3x$$

$$W = \begin{vmatrix} y_1 & y_1' \\ y_2 & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} \cos 3x & -3\sin 3x \\ \sin 3x & 3\cos 3x \end{vmatrix}$$

$$= 3\cos^2 3x + 3\sin^2 3x$$

$$= 3(\cos^2 3x + \sin^2 3x)$$

$$= \underline{\underline{3}}$$

$$y_p(x) = uy_1 + vy_2$$

$$u = - \int \frac{y_2 \cdot \text{RHS}}{W} dx$$

$$= - \int \frac{\sin 3x \cdot \cos 3x}{3} dx$$

$$= -\frac{1}{3} \int \frac{\sin 3x}{\sin 3x} dx$$

$$= -\frac{1}{3} \int 1 dx = -\frac{1}{3}x + C$$

$$v = \int \frac{y_1 \cdot \text{RHS}}{W} dx$$

$$= \int \frac{\cos 3x \cdot \operatorname{cosec} 3x}{3} dx$$

$$= \frac{1}{3} \int \frac{\cos 3x}{\sin 3x} dx$$

$$= \frac{1}{3} \int \cot 3x dx = \frac{1}{9} \log |\sin x| + C$$

$$= \frac{1}{3} \log |\operatorname{cosec} 3x| + x$$

$$y_p(x) = -\frac{x}{3} \cos 3x + \frac{1}{9} \sin 3x \log |\sin 3x|$$

$$y(x) = C_1 \cos 3x + C_2 \sin 3x - \frac{x \cos 3x}{3} + \frac{\sin 3x}{9} \log |\sin 3x|$$

$$5) (D^2 + 2D + 2I)y = 4e^{-x} \sec^3 x$$

$$\lambda^2 + 2\lambda + 2 = 0$$

$$\lambda = -1 \pm i$$

$$y_h(x) = \underline{\underline{e^{-x}(c_1 \cos x + c_2 \sin x)}}$$

$$y_1 = e^{-x} \cos x \quad y_2 = e^{-x} \sin x$$

$$y_p(x) = u y_1 + v y_2$$

$$W = \begin{vmatrix} e^{-x} \cos x & -\sin x e^{-x} - e^{-x} \cos x \\ e^{-x} \sin x & \cos x e^{-x} - e^{-x} \sin x \end{vmatrix}$$

$$= \underline{\underline{e^{-2x}}}$$

$$u = - \int \frac{e^{-x} \sin x \cdot 4 e^{-x} \sec^3 x}{e^{-2x}} dx$$

$$= -4 \int \sin x \cdot \sec^3 x dx$$

$$= -4 \int \frac{\sin x}{\cos^3 x} dx$$

$$= -4 \int \cos^{-3} x \cdot \sin x dx$$

$$= -4 \frac{\cos^{-2} x}{-2} = \frac{2}{\cos^2 x}$$

$$v = - \int \frac{e^{-x} \cos x \cdot 4 e^{-x} \sec^3 x}{e^{-2x}} dx$$

$$= \int e^{-x} \cos x \cdot 4 \frac{\cos x}{\cos^3 x} = 4 \int \sec^2 x$$

$$\left\{ \begin{aligned} \int [f(x)]^n f'(x) dx \\ = \frac{[f(x)]^{n+1}}{n+1} \end{aligned} \right.$$

$$\begin{aligned} \tan^2 x &= \frac{\sin^2 x}{\cos^2 x} \\ &= \frac{1 - \cos^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} - \frac{\cos^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} - 1 \end{aligned}$$

$$= 4 \tan x.$$

$$y(x) = \frac{2}{\cos^2 x} \cdot e^{-x} \cos x + 4 \tan x \cdot e^{-x} \sin x$$

$$= -2 \tan^2 x \cdot e^{-x} \cos x + 4 \tan x \cdot e^{-x} \sin x$$

$$= -2 e^{-x} \left[\frac{\sin^2 x}{\cos x} + \frac{2 \sin^2 x}{\cos x} \right]$$

$$= -2 e^{-x} \left[\frac{3 \sin^2 x}{\cos x} \right]$$

$$= \underline{\underline{-2 e^{-x} [\tan x \cdot \sin x]}}$$

KTUNOTES.IN

$$6) (D^2 - 2D + 1)y = 35 x^{3/2} e^x$$

$$\lambda^2 - 2\lambda + 1 = 0$$

$$\lambda = 1, 1$$

$$y_h(x) = (C_1 + C_2 x) e^x$$

$$y_1 = e^x \quad y_2 = x e^x$$

$$y_p(x) = u y_1 + v y_2$$

$$W = \begin{vmatrix} e^x & e^x \\ x e^x & x e^x + e^x \end{vmatrix} = \underline{\underline{e^{2x}}}$$

$$\begin{aligned} u &= - \int \frac{x e^x - 35 x^{3/2} e^x}{e^{2x}} dx \\ &= -35 \int x^{5/2} dx = -35 \left[\frac{x^{5/2+1}}{5/2} \right] \\ &= \underline{\underline{-10 x^{7/2}}} \end{aligned}$$

$$\begin{aligned} v &= \int \frac{e^x \cdot 35 x^{3/2} e^x}{e^{2x}} dx \Rightarrow 35 \int x^{3/2} dx \\ &= 35 \cdot x^{5/2} \times \frac{2}{5} \\ &= \underline{\underline{14 x^{5/2}}} \end{aligned}$$

$$y_p(x) = -10 x^{7/2} e^x + 14 x^{5/2} x e^x$$

$$y(x) = \underline{(C_1 + C_2 x) e^x - 10 x^{7/2} e^x + 14 x^{5/2} e^x}$$

Method II

Method of Undetermined Coefficients

Consider $y'' + ay' + by = r(x) \longrightarrow \textcircled{1}$

where $r(x)$ can be

- 1) $e^x \rightarrow$ an exponential func.
- 2) $x^n \rightarrow$ where $n = 0, 1, 2, 3$, a power of x
- 3) $\cos wx$
or $\sin wx \rightarrow$
- OR
- 4) $e^{ax} (\cos wx + \sin wx) \rightarrow$

Choice of rules for method of undetermined coefficients.

Basic Rule :-

If $r(x)$ is any one of the above functions choose $y_p(x)$ as same as the $r(x)$ R.H.S & determine the undetermined coefficients by substituting y_p and its derivatives in eq $\textcircled{1}$

Rule II

Modification rule

- * If a term in the choice for y_p happens to be a solution of the homogeneous ODE corresponding to eq ①. Multiply this term by x .

[OR by x^2 , if this soln. corresponds to a double root of the characteristic equation of the homogeneous ODE]

Sum rule

If $r(x)$ is a sum of functions mentioned above choose y_p as the sum of the functions in the RHS.

Table for undetermined Coefficients

Term in $r(x)$	choice in $y_p(x)$
1) e^{ax}	$y_p(x) = k e^{ax}$
2) x^n	$y_p(x) = k_n x^n + k_{n-1} x^{n-1} + \dots + k_1 x + k_0$
3) $\sin \omega x$ or $\cos \omega x$	$y_p(x) = k \cos \omega x + \sum_{n=1}^M k_n \sin \omega x$
4) $e^{ax} [\sin \omega x \cos \omega x + \sin \omega x]$	$y_p(x) = e^{ax} (k \cos \omega x + \sum_{n=1}^M k_n \sin \omega x)$

*) Solve by method of undetermined coefficients.

$$y'' + 5y' + 6y = 2e^{-x}$$

$$y'' + 5y' + 6y = 0$$

$$\lambda^2 + 5\lambda + 6 = 0$$

$$(\lambda + 2)(\lambda + 3) = 0$$

$$\lambda = -2, -3$$

$$y_h(x) = \underline{\underline{C_1 e^{-2x} + C_2 e^{-3x}}} \longrightarrow \textcircled{1}$$

choose $y_p(x) = k e^{-x}$

Since $y_p(x)$ is a particular solution of the given ODE,

$$y_p'(x) = -k e^{-x}$$

$$y_p''(x) = k e^{-x}$$

Substitute the terms in the given ODE

$$y_p''(x) + 5y_p'(x) + 6y_p(x) = 2e^{-x}$$

$$k e^{-x} + 5(-k e^{-x}) + 6k e^{-x} = 2e^{-x}$$

$$2k e^{-x} = 2e^{-x}$$

$$\underline{\underline{k = 1}}$$

$$y_p(x) = e^{-x}$$

$$\underline{y(x) = C_1 e^{-2x} + C_2 e^{-3x} + e^{-x}}$$

Q) Solve the initial value problem $y'' + 3y' + 2.25y = -10e^{-1.5x}$

given that $y(0) = 1$ $y'(0) = 1$

ans) $y'' + 3y' + 2.25y = -10e^{-1.5x}$

$$\lambda^2 + 3\lambda + 2.25 = 0$$

$$\lambda = -3/2, -3/2$$

$$= -1.5, -1.5$$

$$y_h(x) = (C_1 + C_2 x) e^{-1.5x}$$

Choose $y_p(x) = k x^2 e^{-1.5x}$

[Since $e^{-1.5x}$ is repeated here we multiply with x^2]

$$y_p(x) = k (x^2 \cdot (-1.5) e^{-1.5x} + 2x e^{-1.5x})$$

[Since $\lambda = -1.5x$ is double root of the characteristic eq]

$$= -k x$$

$$= -k (-1.5x^2 e^{-1.5x})$$

$$= k [-1.5x^2 + 2x] e^{-1.5x}$$

$$y_p'(x) = k [-1.5 \cdot 2x + 2]$$

$$y''_p(x) = k \left\{ (-1.5x^2 + 2x)(-1.5)e^{-1.5x} + (-3x + 2)e^{-1.5x} \right\}$$

$$= k \left\{ (1.5x^2 - 3x + 2 - 3x) e^{-1.5x} \right\}$$

$$= \underline{k \{ (1.5x^2 - 6x + 2) \} e^{-1.5x}}$$

$$y'_p(x) = k \left[(x^2)(1.5)e^{-1.5x} + 2 \{ e^{-1.5x} \} \right]$$

$$= \underline{k \{ -1.5x^2 + 2x \} e^{-1.5x}}$$

Substitute in ODE,

$$y''_p(x) + 3y'_p(x) + 2.25y_p(x) = -10e^{-1.5x}$$

$$= k \{ (1.5x^2 - 6x + 2) e^{-1.5x} \} + k \{ -1.5x^2 + 2x \} e^{-1.5x} +$$

$$+ 2.25k \{ -1.5x^2 + 2x \} e^{-1.5x}$$

$$2.25 \left[k(x^2 e^{-1.5x}) \right] = -10e^{-1.5x}$$

Compare the coefficients of x^2 , x & constant term in both sides,

$$k[1.5]^2 - 3k(1.5) + 2.25k = 0$$

$$2.25k - 4.5k + 2.25k = 0$$

$$4.5k - 4.5k = 0$$

$$-6k + 6k = 0$$

$$\underline{0 = 0}$$

Compare the constants,

$$2k = -10$$

$$\underline{\underline{k = -5}}$$

$$y_p(x) = -5x^2 e^{-1.5x}$$

$$\underline{\underline{y(x) = (C_1 + C_2 x) e^{-1.5x} + (-5)x^2 e^{-1.5x}}}$$

$$y'(x) = e^{-}$$

$$y(x) = e^{-1.5x} [C_1 + C_2 x - 5x^2] e^{-1.5x}$$

$$y'(x) = C_1 + C_2 x - 5x^2$$

$$(C_1 + C_2 x - 5x^2) (-1.5) e^{-1.5x}$$

$$+ e^{-1.5x} (0 + C_2 - 10x)$$

$$y(0) = 1, \quad y'(0) = 1$$

$$y(0) = [C_1 + C_2(0)(-5)(1)^2] e^{-1.5(0)} = 1$$

$$= \cancel{(C_1 - 5)} e^0 = 1$$

$$= \cancel{5} = C_1 e^0 = y(0)$$

$$\underline{\underline{C_1 = 1}}$$

$$y'(0) = C_1(-1.5) e^0 + C_2 e^0$$

$$\Rightarrow -1.5 + C_2$$

$$\Rightarrow 1.5 + 1 = C_2$$

$$\Rightarrow C_2 = 2.5$$

$$\text{Sol. } \underline{\underline{y(x) = (1 + 2.5x - 5x^2) e^{-1.5x}}}$$

Q) Solve y by method of undetermined coefficients,

$$y'' + 3y' + 2y = 12x^2$$

$$\lambda^2 + 3\lambda + 2 = 0 \quad [\text{characteristic eq}]$$

$$\lambda = -1, -2$$

$$\underline{\underline{y_h(x) = C_1 e^{-x} + C_2 e^{-2x}}}$$

$$\text{choose } y_p(x) = k_2 x^2 + k_1 x + k_0$$

$$y_p'(x) = 2k_2 x + k_1 + 0$$

(because RHS
= $12x^2$)

choose y_p as a
2^o polynomial]

$$y_p''(x) = 2k_2$$

$$y_p''(x) + 3y_p'(x) + 2y_p(x) = 12x^2$$

$$2k_2 + 3(2k_2x + k_1) + 2[k_2x^2 + k_1x + k_0] = 12x^2$$

Compare the coefficients of x^2 , x & constant terms in both sides,

Coefficients of x^2

$$2k_2 = 12$$

$$2k_2 = 12$$

$$k_2 = 6$$

Coefficients of x

$$6k_2 + 2k_1 = 0$$

$$6 \times 6 + 2k_1 = 0$$

$$k_1 = -36/2 = -18$$

for constant terms

$$3k_1 + 2k_0 = 0$$

$$3(18) + 2(k_0) = 0$$

$$k_0 = \frac{3 \times 18}{2}$$

$$= 3 \times 9 = \underline{\underline{27}}$$

$$2k_2 + 3k_1 + 2k_0 = 0$$

$$2(6) + 3(-18) + 2k_0 = 0$$

$$12 + 3(-18) + 2k_0 = 0$$

$$\underline{\underline{k_0 = 21}}$$

$$\therefore y_p(x) = 6x^2 \cdot (-18x) + 21$$

$$y(x) = y_h(x) + y_p(x)$$

$$\underline{\underline{y(x) = C_1 e^{-2x} + C_2 e^{-x} + 6x^2 - 18x + 21}}$$

Solve $y'' - 4y = 8 \cos \pi x$

$$y'' - 4y = 0$$

$$\lambda^2 - 4 = 0$$

$$\lambda^2 = 4$$

$$\lambda = 2, -2$$

$$\underline{\underline{y_h(x) = (C_1 + C_2 x) e^{2x} + C_1 e^{-2x} + C_2 e^{-2x}}}$$

$$y_p(x) = K \cos \pi x + M \sin \pi x$$

$$\begin{aligned} y_p'(x) &= K(-\sin \pi x)\pi + M(\cos \pi x)\pi \\ &= -K\pi \sin \pi x + M\pi \cos \pi x \end{aligned}$$

$$\begin{aligned} y_p''(x) &= -K\pi(\cos \pi x)\pi + M\pi(-\sin \pi x)\pi \\ &= -K\pi^2 \cos \pi x - M\pi^2 \sin \pi x \end{aligned}$$

$$y_p''(x) - 4y_p(x) = 8 \cos \pi x$$

$$\begin{aligned} (-K\pi^2 \cos \pi x - M\pi^2 \sin \pi x) - 4(K \cos \pi x + M \sin \pi x) \\ = 8 \cos \pi x \end{aligned}$$

$$-K\pi^2 \cos \pi x - M\pi^2 \sin \pi x - 4K \cos \pi x - 4M \sin \pi x$$

$$= 8 \cos \pi x$$

$$(-K\pi^2 - 4K) \cos \pi x - M$$

$$\cos \pi x (-K\pi^2 - 4K) + \sin \pi x (-M\pi^2 - 4M)$$

$$= 8 \cos \pi x$$

Comparing coefficients of $\cos \pi x$

$$-K\pi^2 - 4K = 8$$

$$-K(\pi^2 + 4) = 8$$

$$-K = \frac{8}{\pi^2 + 4}$$

comparing coefficients of $\sin \pi x$

$$-M\pi^2 - 4M = 0$$

$$-M(\pi^2 + 4) = 0$$

$$-M = 0$$

$$\underline{M = 0}$$

$$y_p(x) = \frac{-8}{\pi^2 + 4} \cos \pi x + 0 \sin \pi x$$

$$= \frac{-8 \cos \pi x}{\pi^2 + 4}$$

$$\underline{y(x) = (C_1 e^{2x} + C_2 e^{-2x}) + \frac{-8 \cos \pi x}{\pi^2 + 4}}$$

Q5) Solve $y'' + 4y' + 4y = e^{-x} \cdot \cos x$

$$\lambda^2 + 4\lambda + 4 = 0$$

$$\lambda = -2, -2$$

$$\underline{y_h(x) = (C_1 + C_2 x) e^{-2x}}$$

$$y_p(x) = e^{-x} [K \cos x + M \sin x]$$

$$y_p'(x) = e^{-x} [-K \pi \sin \pi x + M \pi \cos \pi x] + [K \cos x + M \sin x] (-e^{-x})$$

$$= e^{-x} [-k \sin x + M \cos x - k \cos x - M \sin x]$$

$$= e^{-x} [(k+M) \sin x + (M-k) \cos x]$$

$$y'' p(x) = e^{-x} [-(k+M) \cos x - (M-k) \sin x] - e^{-x} [- (k+M) \sin x + (M-k) \cos x]$$

$$= e^{-x} [-(k+M) \cos x - (M-k) \sin x + (k+M) \sin x - (M-k) \cos x]$$

$$= e^{-x} [-(k+M) + (M-k)]$$

$$= e^{-x} [(k+M+M-k) \cos x + (-M+k+k+M) \sin x]$$

$$= e^{-x} [(-2M) \cos x + (2k) \sin x]$$

Substituting,

$$e^{-x} [-2M \cos x + 2k \sin x] + 4e^{-x} [-(k+M) \sin x + (M-k) \cos x]$$

$$+ 4e^{-x} [k \cos x + M \sin x] = e^{-x} \cos x$$

Comparing Coefficients of $\cos x$

$$-2M + 4(M-k) + 4k = 1$$

$$-2M + 4M - 4k + 4k = 0$$

$$-2M = 1$$

$$M = (-1/2)$$

$$-2M + 4(M-k) + 4k = 1$$

$$2M = 1$$

$$M = 1/2$$

Equating Coefficients of $\sin x$

$$2k - 4(k+M) + 4M = 0$$

$$2k - 4k - 4M + 4M = 0$$

$$2k - 4k = 0$$

$$k = 0$$

$$y_p(x) = e^{-x} \left[\frac{1}{2} \sin x \right]$$

$$\underline{y(x) = (C_1 + C_2 x) e^{-2x} + e^{-x} \left(\frac{1}{2} \sin x \right)}$$

Solve $(D^2 - 4D + 3I)y = e^x - \frac{9}{2}x$.

$$D^2 \lambda^2 - 4\lambda + 3 = 0$$

$$\lambda = 3, 1$$

$$y_h(x) = \underline{C_1 e^{3x} + C_2 e^x}$$

e^x appears more
times in $y_h(x)$

$$y_p(x) = (k \cdot x \cdot e^x) + (k_1 x + k_0)$$

$$y'_p(x) = \underline{k[x e^x + e^x] + [k_1]} = \cancel{k e^x} = k e^x(x+1) + k_1$$

$$y''_p(x) = k[x e^x + e^x + e^x] + 0$$

$$= \underline{k[x e^x + 2e^x]} = k e^x(1) + k e^x(x+1)$$

Substitute,

$$= k x e^x + 2k e^x - 4[k x e^x + k e^x + k_1] + 3[k x e^x + k_1 x + k_0] = e^x - \frac{9}{2}x$$

Comparing coefficients:-

1-413

$$\underline{x e^x}$$

$$k - 4k + 3k = 0$$

$$\underline{k = 0}$$

$$2k - 4k = 1$$

$$ke^x + ke^x(x+1) - 4[ke^x(x+1) + k_1 + 3[kxe^x + kx + k_0]] = e^x - 9/2 x$$

Comparing Coefficients of e^x

$$k + k(x+1) - 4k(x+1) + 3kx = 1$$

$$2k - 4k = 1$$

$$k = -1/2$$

Comparing coefficients of x

$$3k_1 = -9/2$$

$$k_1 = -9/2 \times 1/3 = -3/2$$

Equating coefficients

$$-4k_1 + 3k_0 = 0$$

$$k_0 = -12/2 \times 1/3 = \underline{\underline{-2}}$$

Solve.

$$(3D^2 + 27I)y = 3\cos x + \cos 3x$$

$$3\lambda^2 + 27 = 0$$

$$3\lambda^2 = -27$$

$$\lambda^2 = -9$$

$$\lambda = \pm 3i$$

$$\underline{y_h(x) = C_1 \cos 3x + C_2 \sin 3x}$$

$$y_p(x) = [k \cos x + M \sin x] + x[k_1 \cos 3x + M_1 \sin 3x]$$

$$\left[\begin{array}{l} \text{Multiplication Rule} \\ \cos 3x \text{ appears once in } y_h(x) \\ \text{So multiply by } x \end{array} \right]$$

$$y_p(x) = k \cos x + M \sin x + x k_1 \cos 3x + x M_1 \sin 3x$$

$$y'_p(x) = -k \sin x + M \cos x + k_1 (x \cdot (-3 \sin 3x) + \cos 3x (1)) + M_1 (x \cdot 3 \cos 3x + \sin 3x (1))$$

$$= -k \sin x + M \cos x + -3x k_1 \sin 3x + k_1 \cos 3x + 3x M_1 \cos 3x + M_1 \sin 3x$$

$$+ M x \cdot 3 \cos 3x + M_1 \sin 3x$$

$$= [-k \sin x + M \cos x] + (-3kx + M_1) \sin 3x + (3Mx + k_1) \cos 3x$$

$$y''_p(x) = [-k \cos x - M \sin x] + (3k_1 x + M_1) 3 \cos 3x + (3k_1) \sin 3x + (3M_1 x + k_1) (-3 \sin x)$$

$$= [-k \cos x - M \sin x] + (-9k_1 x + 3M_1 + 3M_1) \cos 3x + (-3k_1 - 9M_1 x - 3k_1) \sin 3x$$

Substitute in (eq) $3y''_p(x) + 27y_p(x) = 0$

$$3[-k \cos x - M \sin x] + 27(-9k_1 x + 3M_1 + 3M_1) \cos 3x + (-3k_1 - 9M_1 x - 3k_1) \sin 3x$$

$$= (-3k \cos x - 3M \sin x) + (-27k_1 x + 18M_1) \cos 3x +$$

$$(-18k_1 - 27M_1 x) \sin 3x + 27k \cos x + 27M \sin x$$

$$+ 27x k_1 \cos 3x + 27x M_1 \sin 3x$$

$$= 3 \cos x + \cos 3x$$

Ex

Equating Coefficients

by $\cos x$

$$-3k + 24k = 3$$

$$= 24k = 3$$

$$k = \underline{\underline{\frac{1}{8}}}$$

$\sin x$

$$24M - 3M = 0$$

$$24M = 0$$

$$\underline{\underline{M = 0}}$$

$\cos 3x$

$$-24k, x + 18M_1 + 24 = 2k, x = 1$$

$$18M_1 = 1$$

$$M_1 = \underline{\underline{\frac{1}{18}}}$$

$$\sin 3x$$

$$-18k_1 - 27M_1 x + 27x = 0 \quad 27M_1 x = 0$$

$$-18k_1 = 0$$

$$\underline{k_1 = 0}$$

$$\text{Soln :- } y(x) = y_h(x) + y_p(x)$$

$$= C_1 \cos 3x + C_2 \sin 3x + \frac{1}{8} \cos x + \frac{1}{18} x \cdot \sin 3x$$

$$m(n^2 - 16)u = 9.6e^{4x} + 30e^x$$

$$= C_1 \cos 3x + C_2 \sin 3x + \frac{1}{18} x \cdot \sin 3x$$

$$Q) (D^2 - 16)y = 9.6e^{4x} + 30e^x$$

$$\lambda^2 - 16 = 0$$

$$\lambda^2 = 16$$

$$\lambda = \pm 4$$

$$y_h(x) = \underline{C_1 e^{4x} + C_2 e^{-4x}}$$

$$y_p(x) = (k_1 x e^{4x}) + (k_2 e^x) \quad (\text{modification rule})$$

$$\begin{aligned} y_p'(x) &= k_1 [x \cdot 4e^{4x} + e^{4x}] + k_2 e^x \\ &= k_1 (4x + 1) e^{4x} + k_2 e^x \end{aligned}$$

$$y' p(x) = k[4x+1]4e^{4x} + 4e^{4x} + k_2 e^x$$

$$= k[(16x+8)e^{4x}] + k_2 e^x$$

Substitute,

$$y'' p(x) - 16y p(x) = 9.6e^{4x} + 30e^x$$

$$16k_1 x e^{4x} + 8k_1 e^{4x} - k_2 e^x - 16k_1 x e^{4x} - 16k_2 e^x$$

$$= 9.6e^{4x} + 30e^x$$

Comparison of coefficients

$$8k_1 = 9.6 \quad \underline{\underline{k_1 = 1.2}}$$

e^x

$$k_2 - 16k_2 = 30$$

$$-15k_2 = 30 \quad \underline{\underline{k_2 = -2}}$$

$$\underline{\underline{y(x) = C_1 e^{4x} + C_2 e^{-4x} + 1.2x e^{4x} - 2e^x}}$$

Solve $y'' + 4y = 8x^2$ $y(0) = 3$; $y'(0) = 0$

$y'' + 4y = -12 \sin 2x$ $y(0) = 1.8$ $y'(0) = 5$

Soln: $-2 \cos 2x + 2x^2 - 1$ $y(0) = 1.8$ $y'(0) = 5$

$+ 1.8 \cos 2x + \sin 2x + 3x \cos 2x$

~~22y''~~

Higher Order Linear non homogeneous ODE with constant coefficients:

Method of undetermined coefficients

Solve $y''' + 3y'' + 3y' + y = 30e^{-x}$ given that $y(0) = 3$
 $y''(0) = -47$; $y'(0) = -3$

$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$

$\lambda = -1, -1, -1$

$y_h(x) = (C_1 + C_2x + C_3x^2)e^{-x}$

$y_p(x) = kx^3(e^{-x})$

$y_p'(x) = kx^3(e^{-x})$

$= k \cdot [x^3 \cdot (-e^{-x}) + e^{-x} \cdot 3x^2]$

$= k(x^3e^{-x} + 3x^2)e^{-x}$

$$y''' p(x) = \dots$$

$$y_p'' = k(-x^3 + 3x^2)(-e^{-x}) + (-3x^2 + 6x)e^{-x}$$

$$= k \int [x^3 - 6x^2 + 6x] e^{-x}$$

$$y''' p(x) = k[(x^3 - 6x^2 + 6x) - (e^{-x})] +$$

$$[(3x^2 - 12x + 6)e^{-x}]$$

$$= k \int [-x^3 + 9x^2 - 18x + 6] e^{-x}$$

~~not~~

$$\text{Solution } y''' p(x) + 3 y'' p(x) + 3 y' p(x) + y p(x) =$$

$$30 e^{-x}$$

$$= k[-x^3 + 9x^2 - 18x + 6]e^{-x} + 3[k(x^3 - 6x^2 + 6x)]e^{-x} \\ + 3[k(-x^3e^{-x} + 3x^2)]e^{-x} + kx^3(e^{-x}) \\ = 30e^{-x}$$

Equating constants

$$-x^3k + 9kx^2 - 18kx + 6k + 3kx^3 - \cancel{3kx^2} + 18kx \\ + -3kx^2 + 19kx^2 + kx^3 = 30$$

$$6k = 30$$

$$\underline{k = 5}$$

$$\therefore y_p(x) = 5x^3(e^{-x})$$

$$y(x) = y_h(x) + y_p(x)$$

$$= (C_1 + C_2x + C_3x^2)e^{-x} + 5x^3e^{-x}$$

$$\cancel{y(x)} = (\cancel{C_1 + C_2x + C_3x^2})e^{-x} + 5x^3e^{-x}$$

$$y(x) = (C_1 + C_2x + C_3x^2 + 5x^3)e^{-x} \rightarrow \textcircled{1}$$

$$\cancel{y'(x) = (C_1 + C_2x + C_3x^2 + 5x^3)(-e^{-x}) + (C_2 + 2C_3x + 15x^2)} \\ \quad \quad \quad 2C_3x + 15x^2$$

$$y'_p(x) = (c_1 + c_2 x + c_3 x^2 + 5x^3)(-e^{-x}) + (c_2 + 2c_3 x + 1.5x^2)e^{-x}$$

$$= (c_2 - c_1) + (2c_3 - c_2)x + (15 - c_3)x^3 - 5x^3 e^{-x}$$

$$y''(x) = [(c_2 - c_1) + (2c_3 - c_2)x + (15 - c_3)x^2 - 5x^3](-e^{-x}) + (2(c_3 - c_2) + 2(15 - c_3) - 15x^2)e^{-x} \rightarrow (2)$$

Put $y(0)$ in in 13

$$y(0) = c_1 e^0$$

$$\underline{3 = c_1}$$

$$\text{Put } y'(0) = (-3) \text{ in (2)}$$

$$y'(0) = c_2 - c_1$$

$$-3 = c_2 - 3$$

$$\underline{c_2 = 0}$$

$$\text{Put } y''(0) = -47$$

$$y''(0) = (c_2 - c_1) - e^0 + 2(c_3 - c_2)$$

$$-47 = C_1 + 2C_3$$

$$-47 = 3 + 2C_3 \therefore \underline{\underline{C_3 = -25}}$$

Sol:

$$\underline{\underline{y(x) = (3 - 25x^2 + 5x^3)e^{-x}}}$$

Scan to enter site
www.ktunotes.in



Get it on play store
bit.ly/ktunotesapp



The logo for KTUnotes, featuring a stylized yellow and orange circular graphic to the right of the text.
KTUnotes
The learning companion.

 admin@ktunotes.in